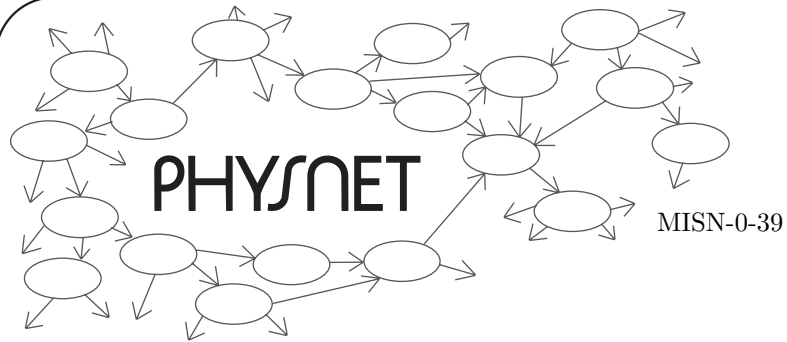
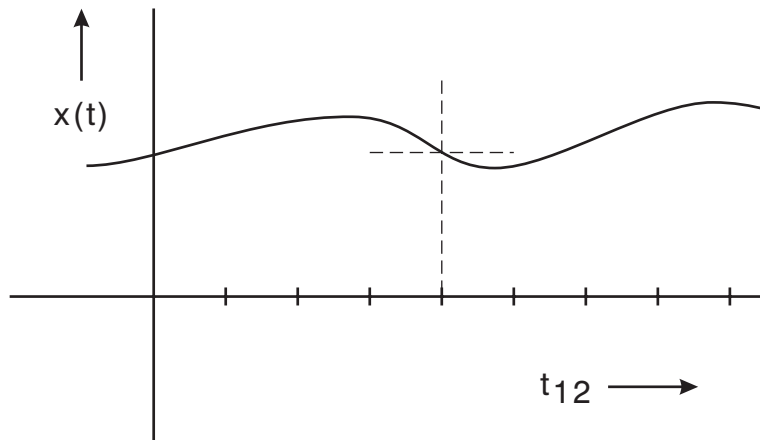




## COMPUTER ALGORITHM FOR THE DAMPED DRIVEN OSCILLATOR



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by  
Peter Signell

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**Input Skills:**

1. Expand a given function about a given point using Taylor's series (MISN-0-4).

**Output Skills (Knowledge):**

- K1. Given the force acting on a damped driven oscillator along with the oscillator's position and velocity at a specified time, derive a Numerov type algorithm for the approximate numerical calculation of the oscillator's position at all past and future times.

**Post-Options:**

1. "Response of a Damped Driven Oscillator" (MISN-0-30).
2. "Damped Driven Oscillations; Mechanical Resonances" (MISN-0-31).
3. "Laplace Transform Solution for the Damped Driven Oscillator" (MISN-0-47).

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New authors, reviewers and field testers are welcome.

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# COMPUTER ALGORITHM FOR THE DAMPED DRIVEN OSCILLATOR

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## 1. Derivation

**1a. Equation for a Damped Driven Oscillator.** The equation to be solved is that of the damped harmonically-driven oscillator<sup>1</sup> acted on by the force:

$$F(t) = -kx(t) - \lambda v(t) + F_0 \cos(\omega t).$$

Since<sup>2</sup>

$$F(t) = ma(t) = m \frac{dx(t)}{dt} \equiv mx''(t),$$

and

$$v(t) = \frac{dx(t)}{dt} \equiv x'(t),$$

our equation to be solved is:

$$mx''(t) + kx(t) + \lambda x'(t) - f(t) = 0, \quad (1)$$

where

$$f(t) \equiv F_0 \cos(\omega t).$$

**1b. Approximate Derivatives by Finite Differences.** We now make two power series expansions<sup>3</sup> about time  $t$ :

$$x(t + \Delta) = x(t) + \Delta x'(t) + (\Delta^2/2)x''(t) + (\Delta^3/6)x'''(t) + \dots$$

$$x(t - \Delta) = x(t) - \Delta x'(t) + (\Delta^2/2)x''(t) - (\Delta^3/6)x'''(t) + \dots$$

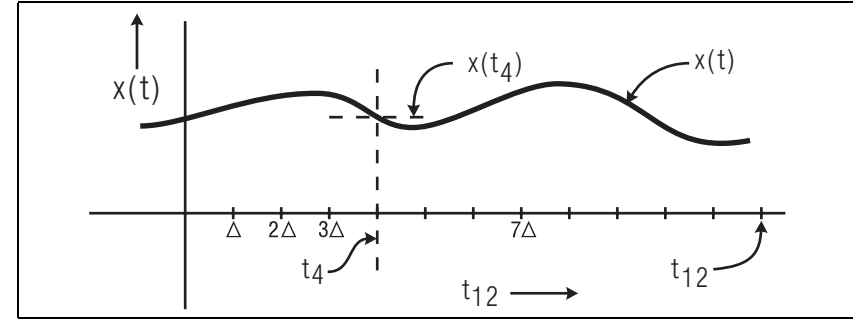
We will choose  $\Delta$  sufficiently small so that we can disregard all terms after  $\Delta^2$  without incurring much error. Then add and subtract the above equations to obtain:

$$x(t + \Delta) + x(t - \Delta) \simeq 2x(t) + \Delta^2 x''(t)$$

<sup>1</sup>See "Damped Driven Oscillations; Mechanical Resonances" (MISN-0-31).

<sup>2</sup>See "Particle Dynamics - The Laws of Motion" (MISN-0-14).

<sup>3</sup>See "Taylor's Series for the Expansion of a Function About a Point" (MISN-0-4).



**Figure 1.** The labeling system for time net-points.

$$x(t + \Delta) - x(t - \Delta) \simeq 2\Delta x'(t)$$

or

$$x''(t) \simeq \frac{1}{\Delta^2} [x(t + \Delta) - 2x(t) + x(t - \Delta)]$$

$$x'(t) \simeq \frac{1}{2\Delta} [x(t + \Delta) - x(t - \Delta)]$$

which are often quoted in calculus courses.<sup>4</sup>

**1c. Net-Point Times.** We need a more succinct labeling system at this point or we won't be able to see the forest for the trees. We define discrete "net-point" times as:  $t_n \equiv n\Delta$ , where  $\Delta$  is some small time interval (see Fig. 1). We can then rewrite the derivatives at time  $t_n$  as:

$$x''_n \simeq \frac{1}{\Delta^2} (x_{n+1} - 2x_n + x_{n-1}) \quad (2)$$

$$x'_n \simeq \frac{1}{2\Delta} (x_{n+1} - x_{n-1}) \quad (3)$$

Then at time  $t_n$  our basic Eq. (1) becomes:

$$mx''_n + kx_n + \lambda x'_n - f_n = 0.$$

Substituting for  $x''_n$  and  $x'_n$ :

$$\frac{m}{\Delta^2} (x_{n+1} - 2x_n + x_{n-1}) + kx_n + \frac{\lambda}{2\Delta} (x_{n+1} - x_{n-1}) - f_n = 0$$

<sup>4</sup>The second equation above is used in "Response of a Damped Driven Oscillator" (MISN-0-30) where a microcomputer is utilized in finding solutions.

Collecting coefficients of the  $x$  values, we get:

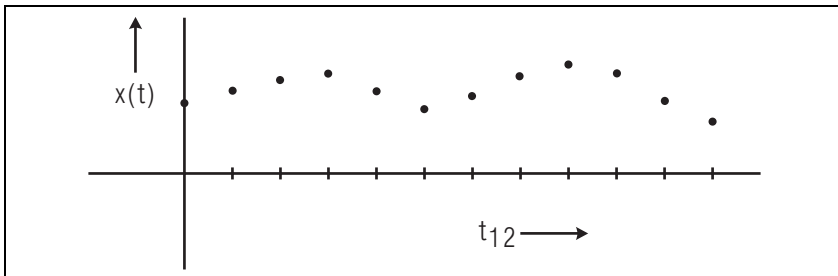
$$x_{n+1} = Ax_n + Bx_{n-1} + \Delta^2 C_n \quad (4)$$

where:

$$\begin{aligned} A &= \left(2 - \Delta^2 \frac{k}{m}\right) / D \\ B &= -\left(1 - \Delta \frac{\lambda}{2m}\right) / D \\ C_n &= \left(\frac{f_n}{m}\right) / D \\ D &= 1 + \Delta \frac{\lambda}{2m} \end{aligned}$$

## 2. Application

**2a. Iterating the Recurrence Relation.** Equation (4) is called a three point recurrence relation because it relates  $x$  at three successive time net-points. If you know  $x_0$  and  $x_1$ , you can use this relation to calculate  $x_2$ . Then knowing  $x_1$  and  $x_2$ , you can calculate  $x_3$ , etc. You can keep up this recurrence procedure until you have reached the time for which you wish to know the solution. For example, this might be a time sufficiently large so that transient effects have died away and only the steady state solution remains.



**Figure 2.** The Calculated Curve. Draw a smooth curve through the plotted points and you have  $x(t)$ .

**2b. Starting the Iteration.** With a three-point recurrence relation one can calculate values at successive time net-points, but one needs two adjacent values in order to get started. In the example cited in the preceding paragraph, one needed  $x_0$  and  $x_1$  in order to start the procedure. Those two initial values generally come from a specification of a “complete set of initial conditions.” Our Eq. (1), being a second order equation for  $x$ , always requires that one specify two independent conditions on  $x$ .

A common case is where both the position and velocity are known at some instant of time. Here we will call that time  $t = 0$ ; hence we know  $x_0$  and  $x_1$  by solving Eq. (3) at  $n = 0$  for  $x_{-1}$  (Note:  $v_0 = x'_0$ ), and using that to eliminate  $x_{-1}$  from Eq. (4) for  $n = 0$ . We thus obtain  $x_1$  in terms of  $x_0$  and  $v_0$  (usually given quantities):

$$x_1 = \left(1 - \Delta^2 \frac{k}{2m}\right) x_0 + \left(\Delta - \Delta^2 \frac{\lambda}{2m}\right) v_0 + \Delta^2 \frac{f_0}{2m}.$$

Now the algorithm is complete. Put in  $x_0$  and  $v_0$  to get  $x_0$  and  $x_1$ ,  $x_0$  and  $x_1$  to get  $x_2$ ,  $x_1$  and  $x_2$  to get  $x_3$ , etc.

▷ How would you obtain position values for times earlier than the one for which  $x$  and  $v$  are initially known? Answer: To see the answer, replace each of the following letters by its successor in the alphabet: mdfzshud cdksz.

## Acknowledgments

Preparation of this module was supported in part by the National Science Foundation, Division of Science Education Development and Research, through Grant #SED 74-20088 to Michigan State University.

## MODEL EXAM

1. The force acting on a damped driven oscillator is  $F = -kx - \lambda v + F_0 \cos(\omega t)$ ; the oscillator's initial position and speed are  $x_0$  and  $v_0$ . Derive without notes, a complete Numerov-type algorithm for the (approximate) calculation of the oscillator's position at all times. NOTE: The algorithm is to include the method of use of any three point recurrence relation such as:

$$x_{n+1} = Ax_n + Bx_{n-1} + \Delta^2 C_n$$

where:

$$\begin{aligned} A &= \left(2 - \Delta^2 \frac{k}{m}\right) / D \\ B &= -\left(1 - \Delta \frac{\lambda}{2m}\right) / D \\ C_n &= \left(\frac{f_n}{m}\right) / D \\ D &= 1 + \Delta \frac{\lambda}{2m} \end{aligned}$$

and boundary conditions such as:

$$x_1 = \left(1 - \Delta^2 \frac{k}{2m}\right) x_0 + \left(\Delta - \Delta^2 \frac{\lambda}{2m}\right) v_0 + \Delta^2 \frac{f_0}{2m}.$$

### Brief Answers:

1. See this module's *text*.

